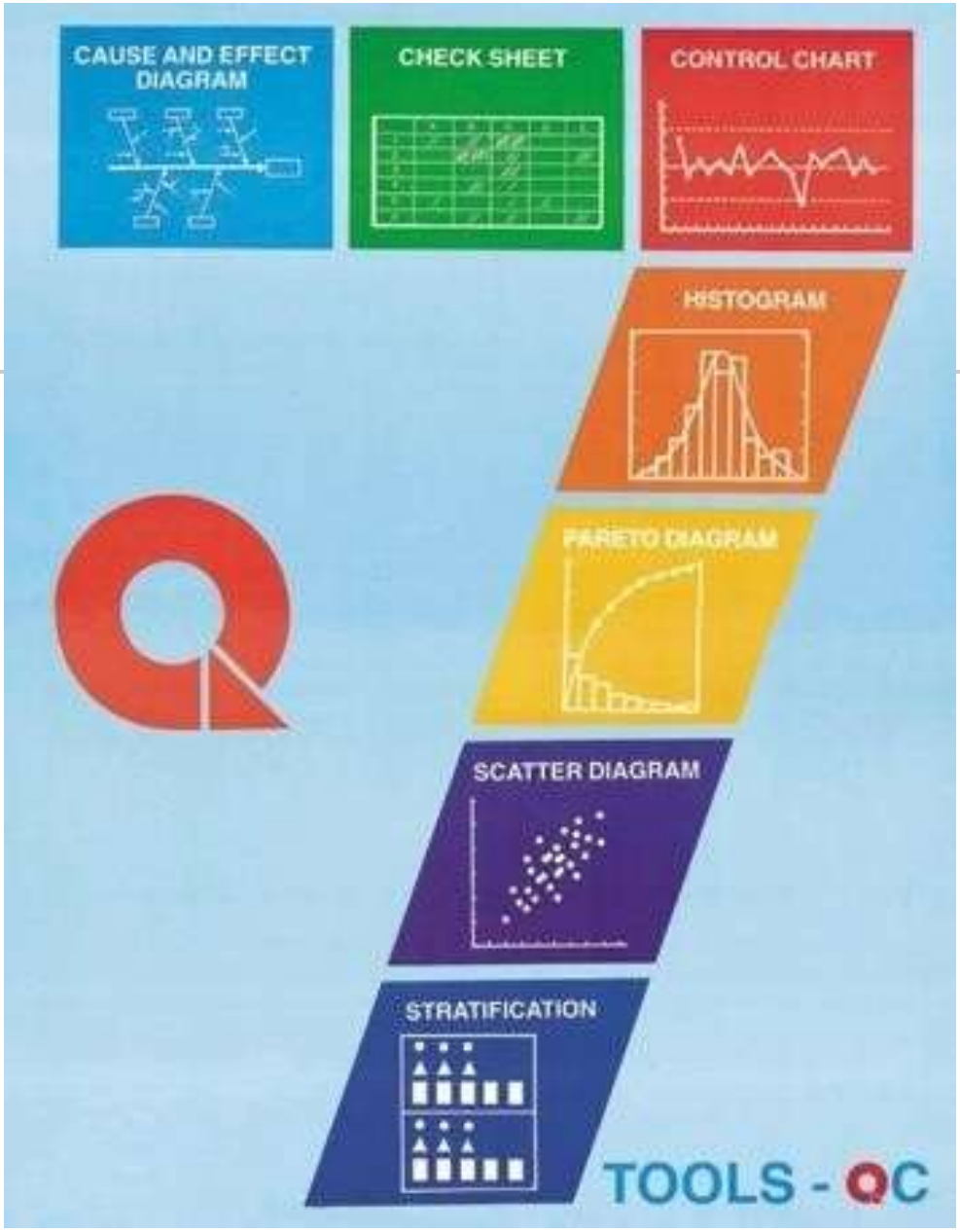


Problem Solving Through 7 QC TOOLS

Module – 3

Total Quality Management

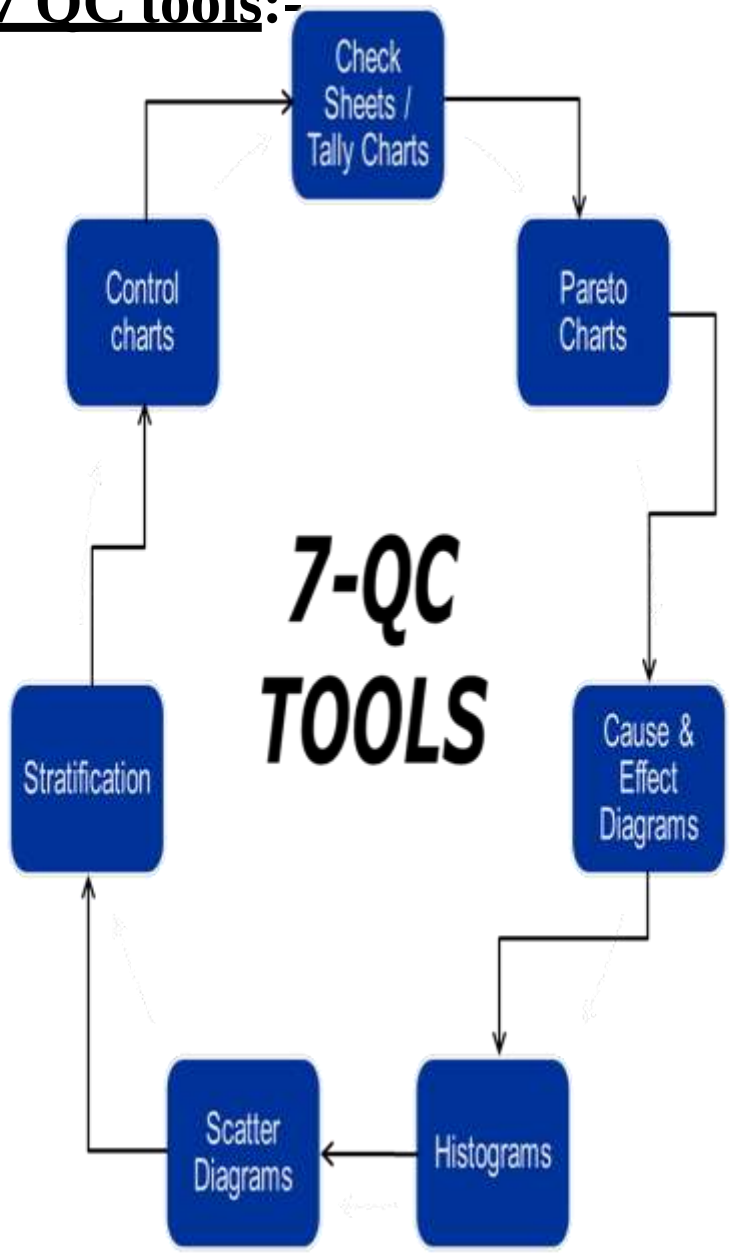
Learning Objectives

- Present an overview of Seven Quality Tools
- Address purpose of each QC tools
- Address application in problem solving
- benefits of each tool

-:Background and Importance of 7 QC tools:-

The **7 QC tools** are simple **statistical tools used for problem solving**. These tools were either **developed in Japan** or introduced to Japan by the Quality Gurus such as **Deming** and **Juran**. **Kaoru Ishikawa** has stated that these 7 tools can be used to solve 95% of all problems.

The 7 Tools of Quality is a designation given to a fixed set of graphical techniques identified as being most helpful in troubleshooting issues related to quality. They are **used to analyze the production process, identify the major problems, control fluctuations of product quality, and provide solutions to avoid future defects.**



Why 7QC Tools?

The Deming Chain



Improve Quality

Decrease Costs

Improve Productivity

Decrease Price

Increase Market

Stay in Business

Provide More Jobs

Return on Investment

-: 7 Quality Control Tools:-

1. Check sheets
2. Stratification
3. Pareto chart
4. Cause and effect diagram
5. Histogram
6. Control chart
7. Scatter diagram





-: PDCA APPROACH :-

PLAN

- **WHAT** → **Definition of problem**
- **WHY** → **Analysis of problem**
- **HOW** → **Identification of causes**

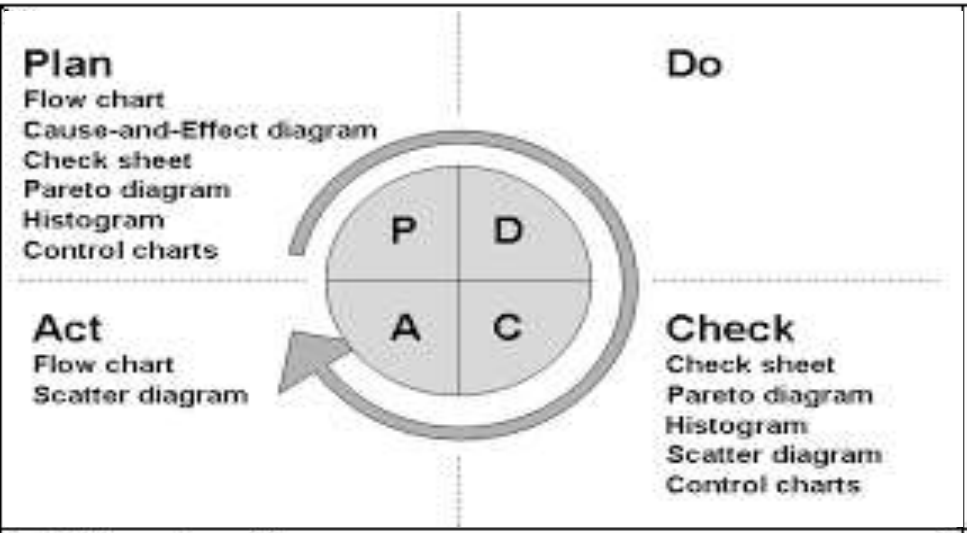
Planning countermeasure

DO → **Implementation**

CHECK → **Confirming effectiveness**

ACT → **Standardizations**

-: PDCA Vs 7-QC TOOLS :-

NO	PDCA	Plan / Do / Check / Act		Pareto Diagram	Cause and effect diagram	Graphs	Check sheets	Histograms	Scatter Diagrams	Control Charts
										
1	P	Definition of problem		@	0	0	@	0		0
2	P	Analysis of problem	Understand current situation	0	0		0	0		0
	P		Set targets	0				0		0
3	P	Identification of cause	Investigate causes & effects		@				0	
	P		Investigate past and current situation	0		@	@	@		@
	P		Try stratification	0	0	0	0	@	@	@
	P		Look at changes over time			0				@
	P		Look at interrelationships	0		0			@	
4	P	Planing Countermeasure				@				
5	D	Implementations			@	0				
6	C	Conforming effectiveness		0		0	0	@	@	@
7	A	Standardisations and establishing control				0	@	0		@

Note : @ Particulary effective technique ; O Effective technique

-: 1st Quality tool: Check Sheet :-

Source of Incoming Model Home Traffic
Check Sheet

	Monday	Tuesday	Wednesday	Thursday	Friday	Total
Daily Newspaper Ads	𝍸 ||	𝍸	𝍸	𝍸	𝍸	24
Weekly Newspaper Ads	|||		||	|	||	8
Website	|||	𝍸	𝍸 |	𝍸 |	||	20
Road Signage		||		|		3
Referral	|				|	2
Total	14	11	13	9	10	57

-: Check Sheet :-

Purpose:-

- Tool for **collecting and organizing** measured or counted data
- Data collected can be used as **input data** for other quality tools
- ➢ Data Collections are based on answering the questions of **What, Where, Who and How**

When to Use a Check Sheet?

-To **collect data repeatedly** by the same person or at the same location.

-To **collect data** on the frequency or patterns of events, problems, defects, defect location, defect causes, etc.

-To **collect data** from a production process.

Type of Defect	Count	Score
Dirty	𝍸 𝍸 ||	12
Broken stitching	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 ||	42
Inconsistent margin	𝍸 𝍸 𝍸	15
Wrinkle	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸	30
Long thread	𝍸 𝍸	10
Padding shape	𝍸 |||	8
Off center	𝍸 𝍸 𝍸 |||	18
Stitch per inch	𝍸 𝍸 𝍸 𝍸 ||||	24
Others	𝍸 𝍸 𝍸 𝍸 ||	22
	Total Defects:	181

-: Check Sheet :-

-: Check Sheet Procedure :-

-Decide what event or problem will be observed. Develop operational definitions.
-Decide when data will be collected and for how long.
-Design the form. Set it up so that data can be recorded simply by making check marks or Xs or similar symbols and so that data do not have to be recopied for analysis. Label all spaces on the form.
-Test the check sheet for a short trial period to be sure it collects the appropriate data and is easy to use. Each time the targeted event or problem occurs, record data on the check sheet.

1 Check Sheet.xlsx



	Days of Week						
Errors	1	2	3	4	5	6	Total
Defective Pilot Light	𝍸 𝍸	𝍸	𝍸	𝍸 III	𝍸 II	𝍸	40
Loose Fasteners	I	III		𝍸	𝍸	II	16
Scratches	I	III	III	I	III	𝍸 𝍸	21
Missing Parts		I		I	I		3
Dirty Contacts	𝍸 I	IIII	𝍸 III	IIII	𝍸	𝍸	32
Other	I	III			II	III	9
Total	19	19	16	19	23	25	121

-: Check Sheet :-

Benefits:

- Collect data in a **systematic** and **organized** manner
- To determine **source** of problem
- To **facilitate** classification of data (stratification)
- The check sheet is a **simple** and **effective way** to display data.
- It provides a **uniform data collection**

Type of Defect	Count	Score
Dirty	𝍸 𝍸 ||	12
Broken stitching	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 ||	42
Inconsistent margin	𝍸 𝍸 𝍸	15
Wrinkle	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸	30
Long thread	𝍸 𝍸	10
Padding shape	𝍸 |||	8
Off center	𝍸 𝍸 𝍸 |||	18
Stitch per inch	𝍸 𝍸 𝍸 𝍸 ||||	24
Others	𝍸 𝍸 𝍸 𝍸 ||	22
	Total Defects:	181

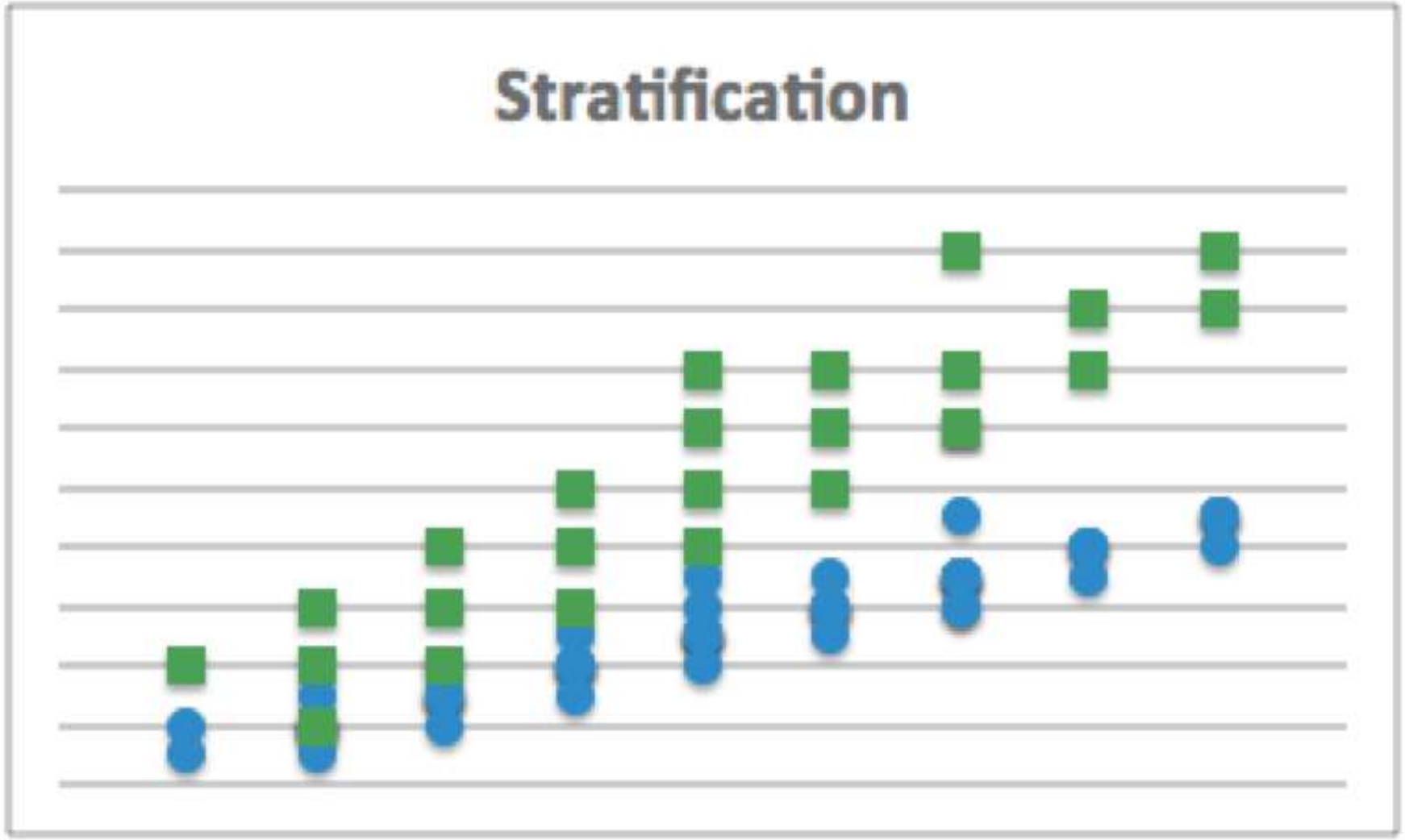
VALUE GENERATION PARTNERS **Check Sheet**

Project Name:

Project Manager: Date:

Defect/Issue	Sun	Mon	Tues	Wed	Thurs	Fri	Sat	Total
								0
								0
								0
								0
								0
								0
								0
Total	0	0	0	0	0	0	0	0

-: 2nd Quality Tool: Stratification :-



-: Stratification :-

Definition:-

Stratification is a system of **formation of layers, classes, or categories**.

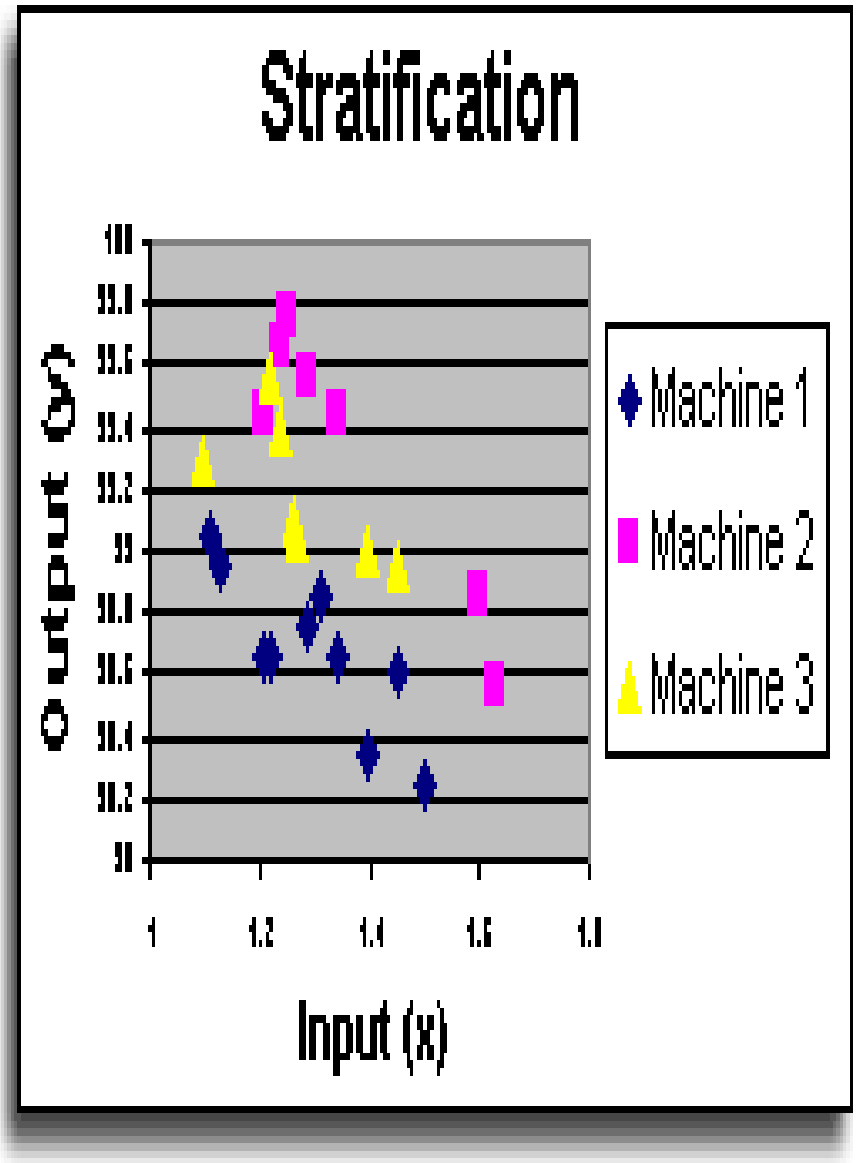
Data collected using check sheets need to be meaningfully classified. Such classification helps **gaining a preliminary understanding** of **relevance and dispersion** of data so that further analysis can be planned to obtain a meaningful output. Meaningful classification of data is called stratification.

This technique separates the data so that patterns can be seen.

When to Use Stratification?

-When **data come from several sources** or **conditions**, such as shifts, days of the week, suppliers or population groups.

- When **data analysis** may require separating different sources or conditions.



-: Stratification :-

Stratification Procedure :-

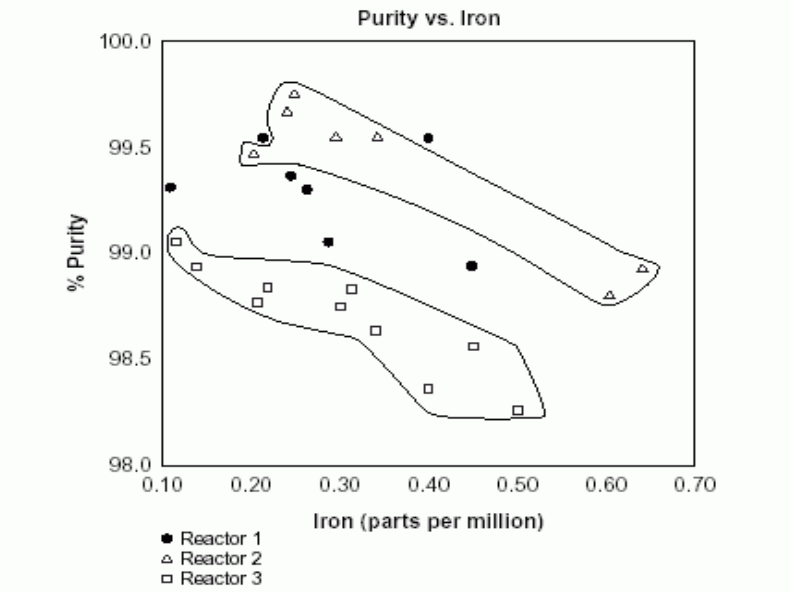
-Before collecting data, consider **which information** about the sources of the data might have an **effect on the results**. **Set up the data collection** so that you collect that information as well.

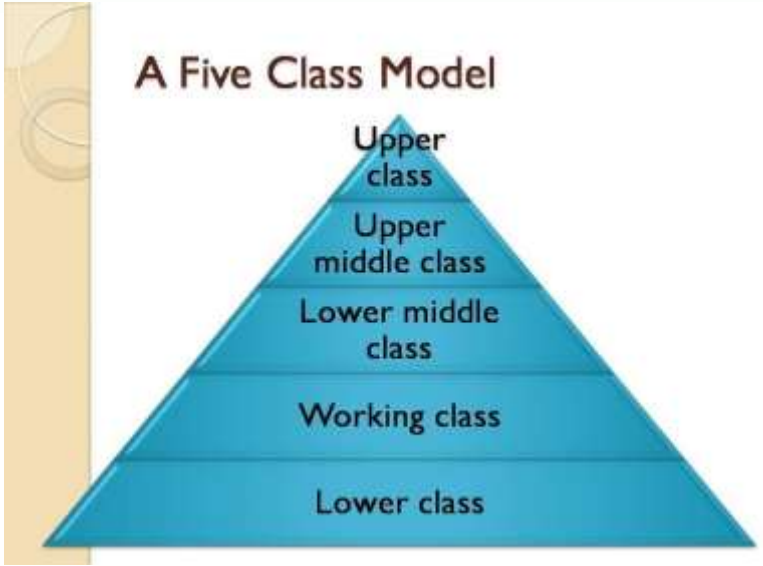
-When plotting or graphing the collected data on a scatter diagram, control chart, histogram or other analysis tool, use different marks or colors to distinguish data from various sources. Data that are **distinguished** in this way are said to be "**stratified**."

-**Analyze** the subsets of stratified data separately.

-**Example:** 1) Variation of object in two different machines 2) Age stratification of two different country 3) Division of society, etc.

2 Stratification diagram.xlsx



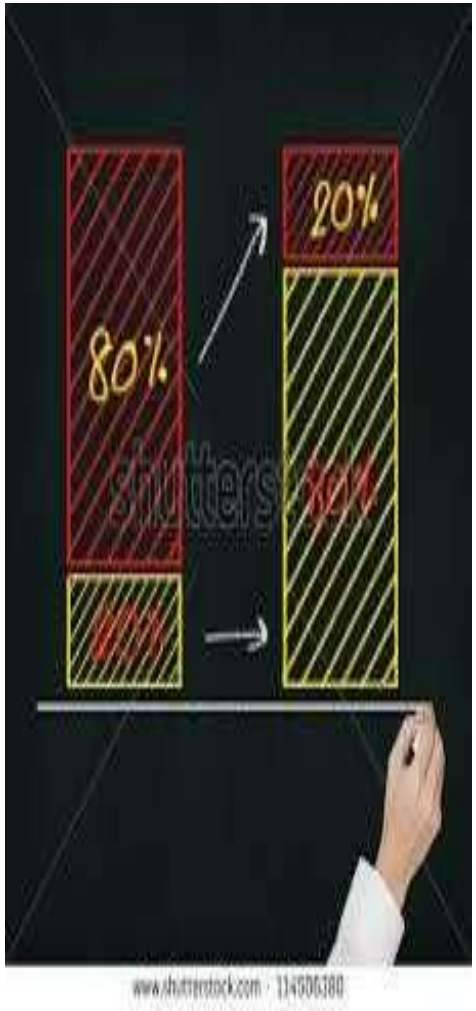


-: 3 Quality tool: Pareto Chart:-



-: Pareto Principle :-

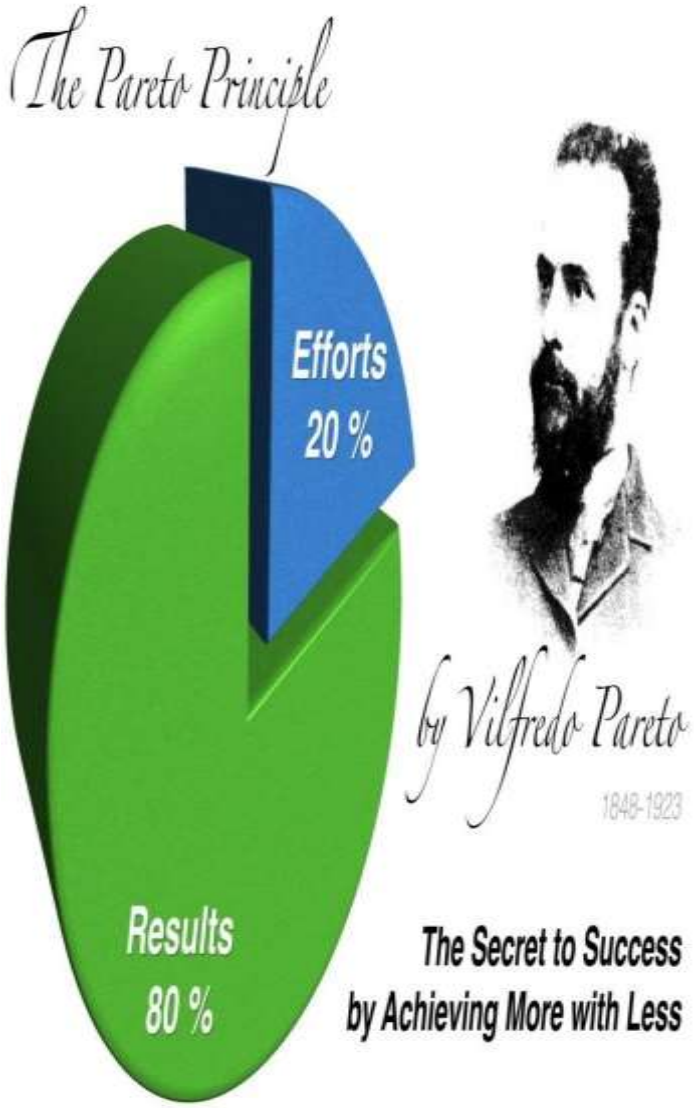
- **Vilfredo Pareto** (1848-1923) Italian economist developed this principle.
  - 20% of the population has 80% of the wealth
- Juran used the term "**vital few, trivial many**". He noted that 20% of the quality problems caused 80% of the dollar loss.
- **Purpose:** The purpose of a Pareto diagram is to separate the significant aspects of a problem from the trivial ones.



7 Quality Tools

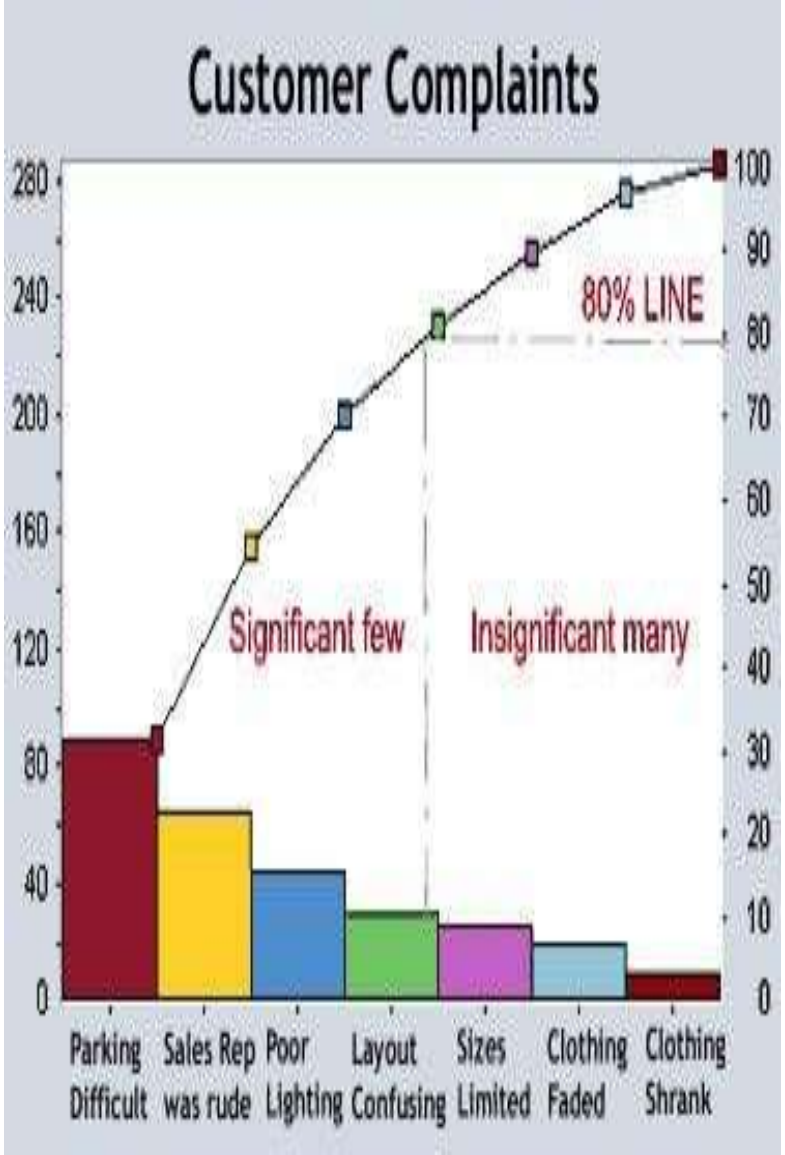
-:Use of Pareto Chart :-

- Pareto charts help teams **focus on the small number** of <u>really important problems</u> or their causes.
- They are useful for **establishing priorities** by showing which are the <u>most critical problems</u> to be tackled or causes to be addressed.
- Pareto chart helps teams to **focus their efforts** where they can have the <u>greatest potential impact</u>.
- When **communicating** with others about your data.



-: Pareto Chart Procedure :-

1. **Develop a list** of problems, items or causes to be compared.
2. **Collect the data** as per defined time frequency
3. **Tally**, for each item, how often it occurred. Determine the **grand total** for all items.
4. **Find the percent** of each item.
5. **List the items** being compared in **decreasing order** of measure of comparison: **e.g.**, the most frequent to the least frequent. The cumulative % for an item is the sum of that item's percent of the total and that of all the other items that come before it in the ordering by rank.



-: Pareto Chart Benefit :-

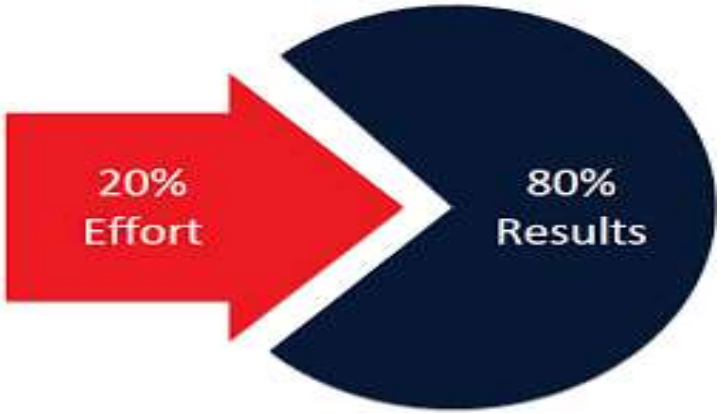
6. List the items on the horizontal axis of a graph from **highest to lowest**. Label the left vertical axis with the numbers, then label the right vertical axis with the cumulative% (the **cumulative total should equal 100%**). Draw in the bars for each item.
7. **Draw a line graph** of the cumulative %. The first point on the line graph should line up with the top of the first bar.
8. Analyze the diagram by identifying most critical items



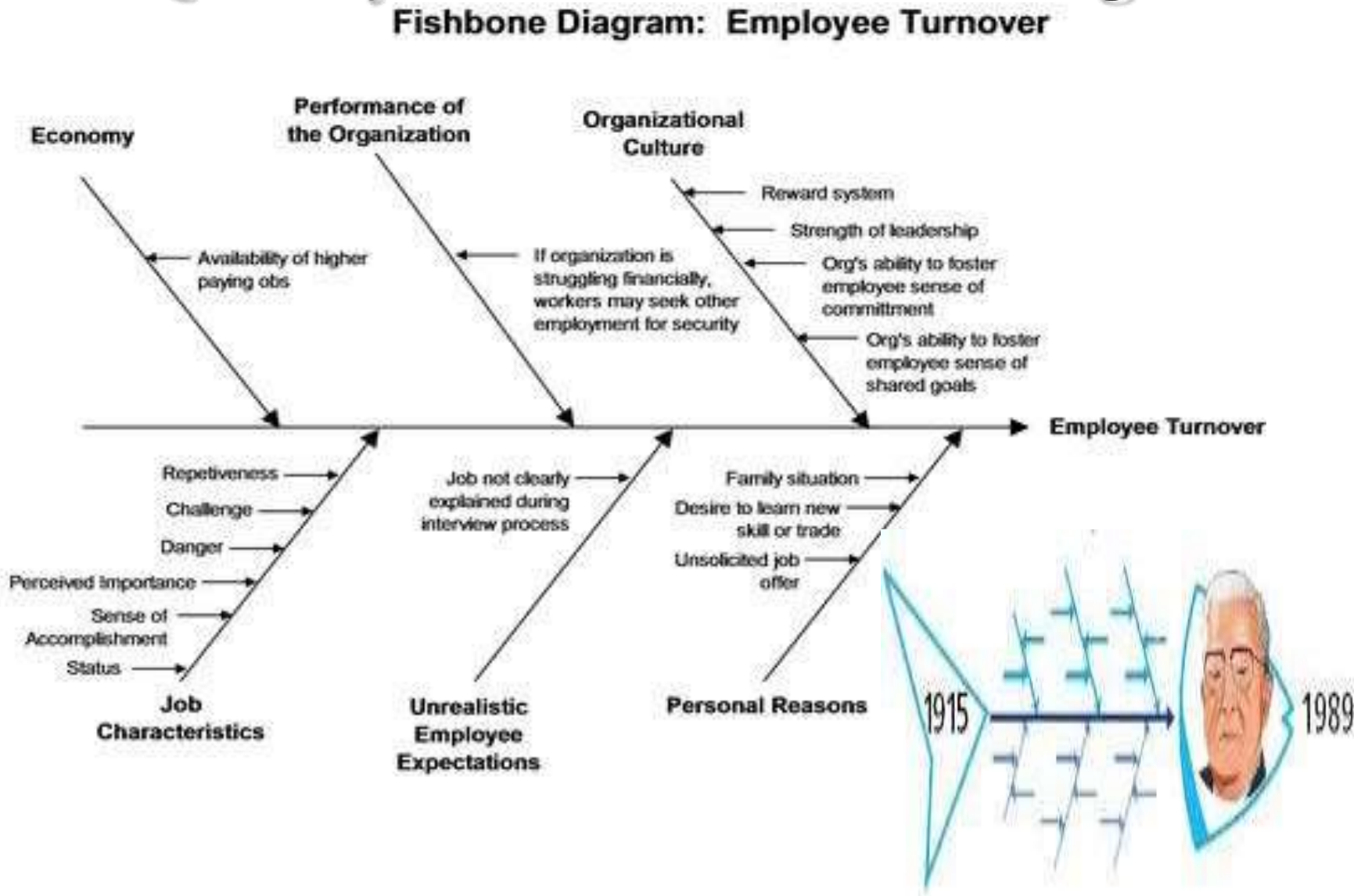
[3 Pareto chart.xlsx]

Benefits:

- ✓ Pareto analysis helps **graphically display results** so the significant few problems emerge from the general background
- ✓ It tells you **what to work on first**



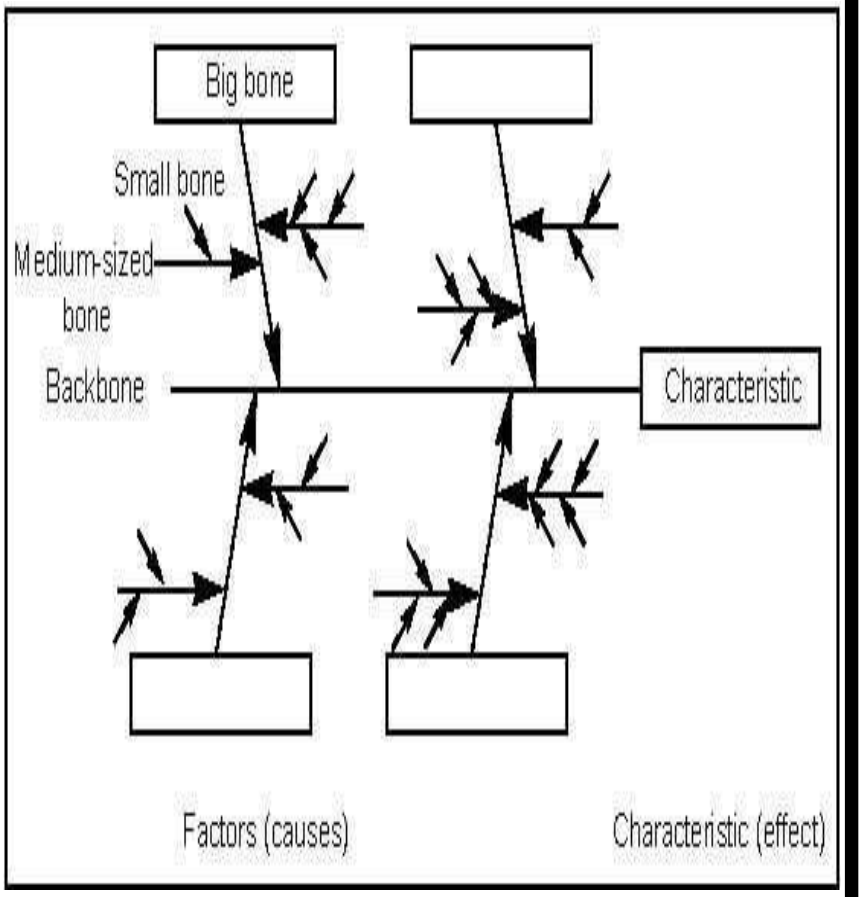
-: 4 Quality Tool: Fishbone diagram :-



-: Fishbone diagram :-

The cause and effect diagram analysis was first developed by ***Professor Kaoru Ishikawa*** of the University of Tokyo in the **1940**s', is also known as the '**Fishbone Diagram**' or the '**Ishikawa Diagram**' or the '**Cause-and-Effect Diagram**'.

Description - The fishbone diagram **identifies many possible causes** for an effect or problem. It can be used to structure a brainstorming session. It immediately sorts ideas into useful categories.



When to use a Fishbone Diagram?

- **When identifying possible causes** for a problem. Especially when a team's thinking tends to fall into ruts.

Fishbone Diagram Procedure -

1)Brainstorm the major categories of causes of the problem.

2) It can be identify by '**6M' techniques**:

i) Methods
ii) Machines (Equipment)
iii) Manpower (People)
iv) Materials
v) Measurement
vi) Management, Environment… etc,

-: Fishbone diagram :-

3)Write the categories of causes as branches from the main arrow.
4)When you are brainstorming causes, consider having team members write each cause on sticky notes, going around the group asking each person for one cause. Ask: "**Why does this happen**?" Continue going through the rounds, getting more causes, until all ideas are exhausted. Causes can be written in several places if they relate to several categories.
5)Analyze causes and eliminate trivial and/or frivolous ideas.
6) Rank causes and **circle** the most likely ones for further consideration and study.
7) Investigate the circled causes.

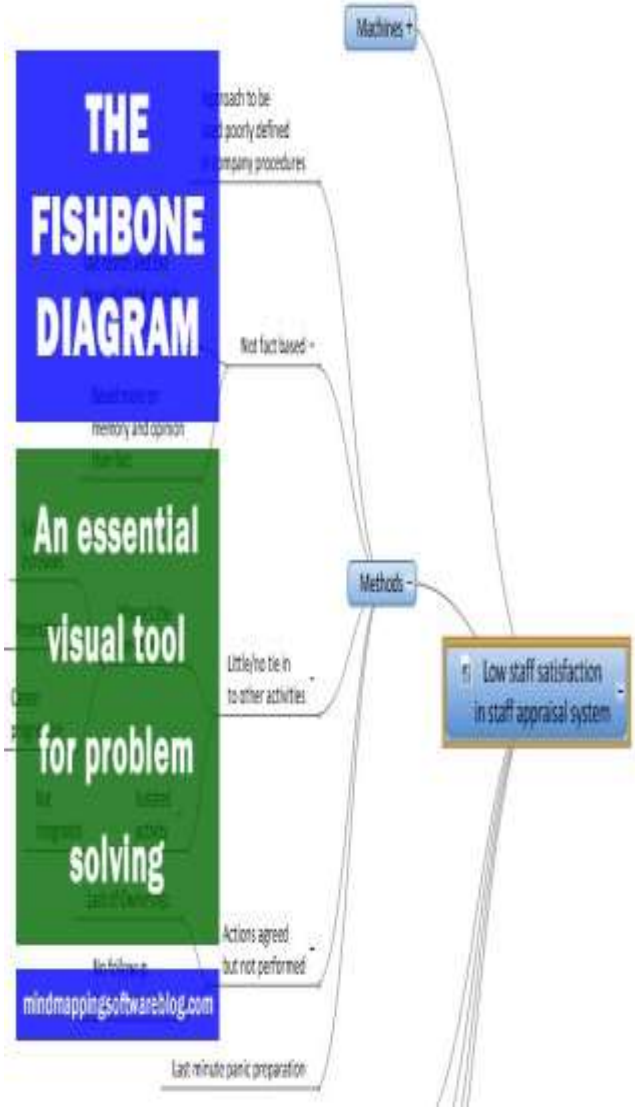


4 Fishbone diagram.xlsx

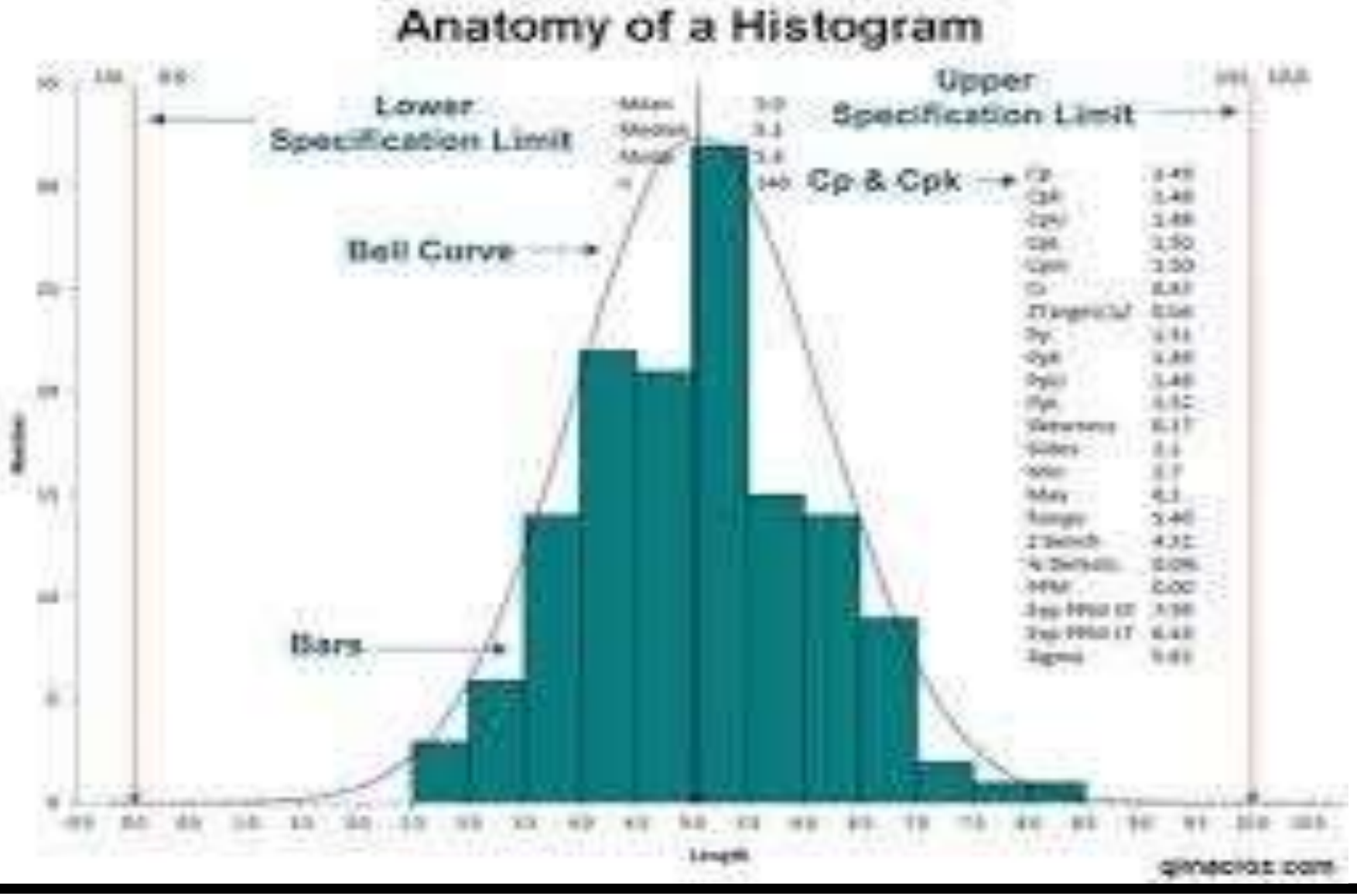
-: Fishbone Diagram :-

Benefits:

- **Breaks problems down** into bite-size pieces to find root cause
- **Fosters/Encourage** team work/participation
- **Common understanding** of factors causing the problem
- **Road map** to verify picture of the process
- **Follows** brainstorming relationship
- **Indicates** possible causes of variation
- **Increases** process knowledge
- Diagram **demonstrates knowledge** of problem solving team
- Diagram is a **guide** for data collection



-: 5 Quality tool: Histogram :-

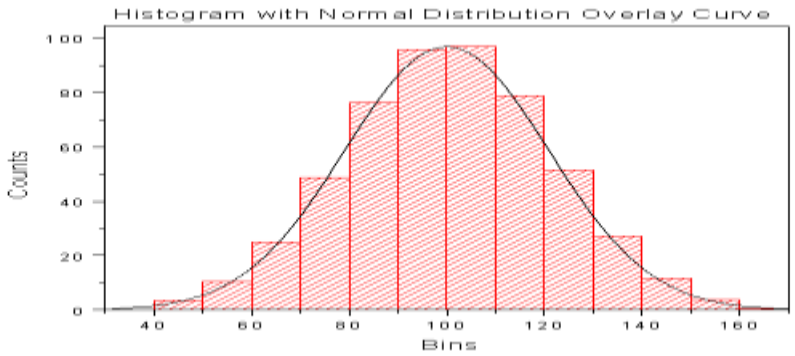


-: Histogram :-

• **Description** -

Histograms are graphs of a **distribution of data** designed to show **centering, dispersion (spread), and shape (relative frequency)** of the data.

They are used to understand how the **output of a process** relates to customer expectations (targets and specifications), and help answer the question: "**Is the process capable of meeting customer requirements?**"



• **When to Use a Histogram?**

1) When the **data are numerical**.
2) When you want to see the **shape of data's distribution**
3) Whether the **output of a process** is distributed approximately normally.
4) When analyzing whether a **process can meet** the customer's requirements.
5) When analyzing what the **output** from a supplier's process looks like.
6)When seeing whether a **process change** has occurred from one time period to another.
7) When determining whether the **outputs** of two or more processes are different.
8) When you wish to **communicate the distribution of data** quickly and easily to others.

-: Histogram :-

Constructing a Histogram

Step 1 - Count number of data points

Step 2 - Summarize on a tally sheet

Step 3 - Compute the range

Step 4 - Determine number of intervals

Step 5 - Compute interval width

-: Histogram :-

Step 6 - Determine interval starting points

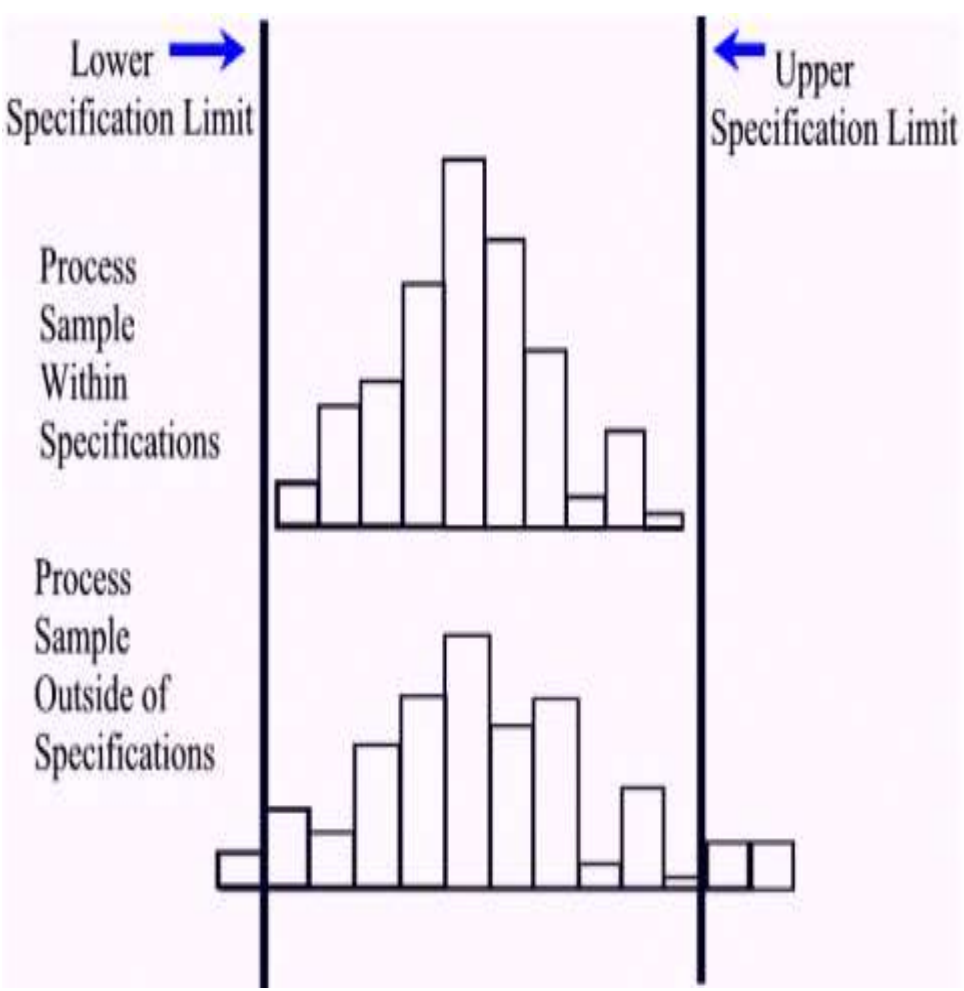
Step 7 - Count number of points in each interval

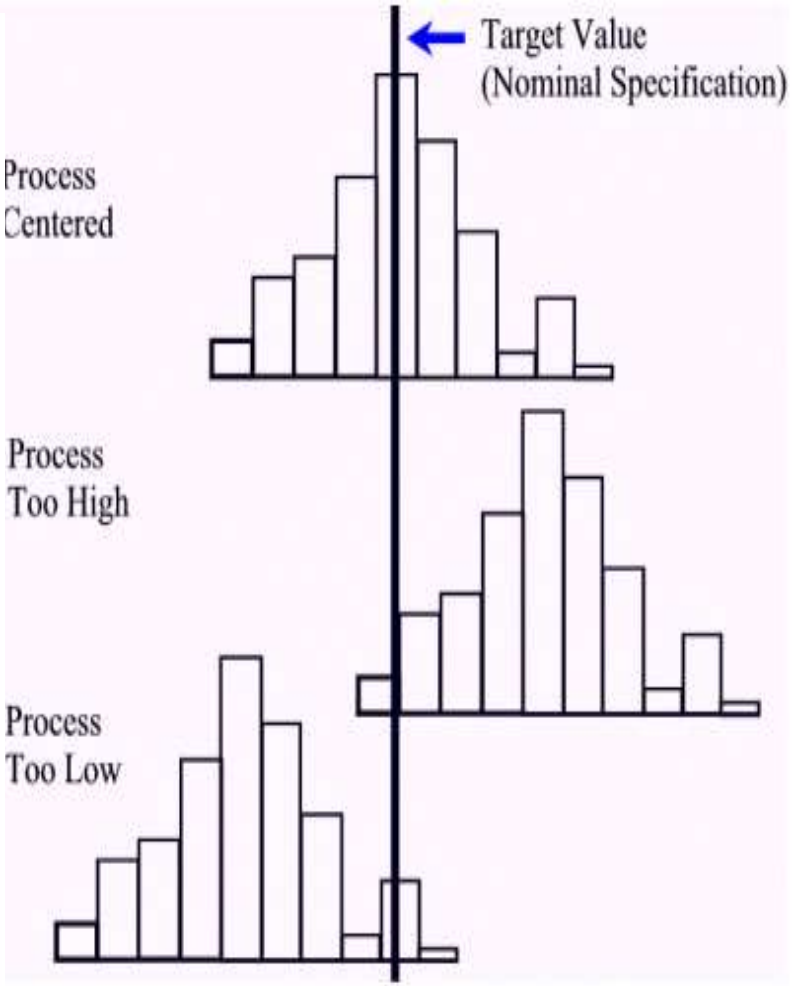
Step 8 - Plot the data

Step 9 - Add title and legend

5 Histogram.xlsx

-: Histogram :-





-: Interpretation of Histogram :-

-: Histogram :-

Definitions :-

C_p= **Process Capability**. A simple and straightforward indicator of process capability.

C_{pk}= **Process Capability Index**. Adjustment of C_p for the effect of non-centered distribution.

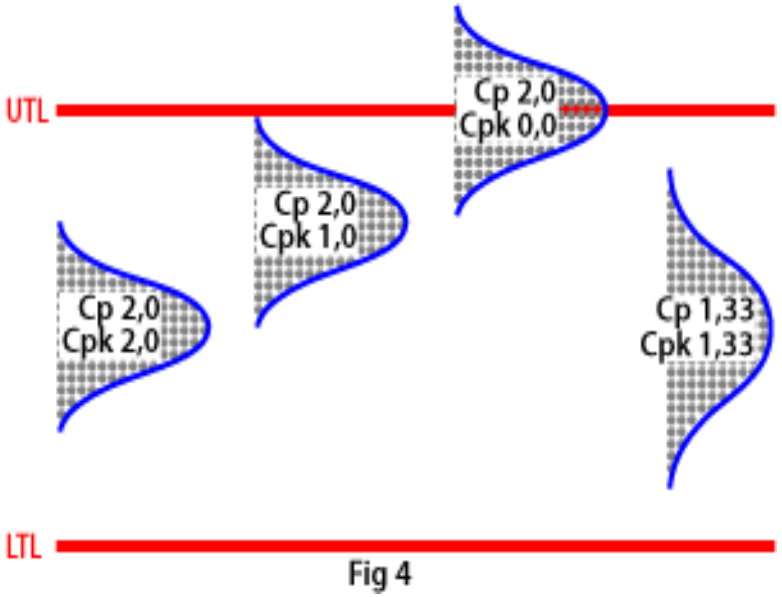


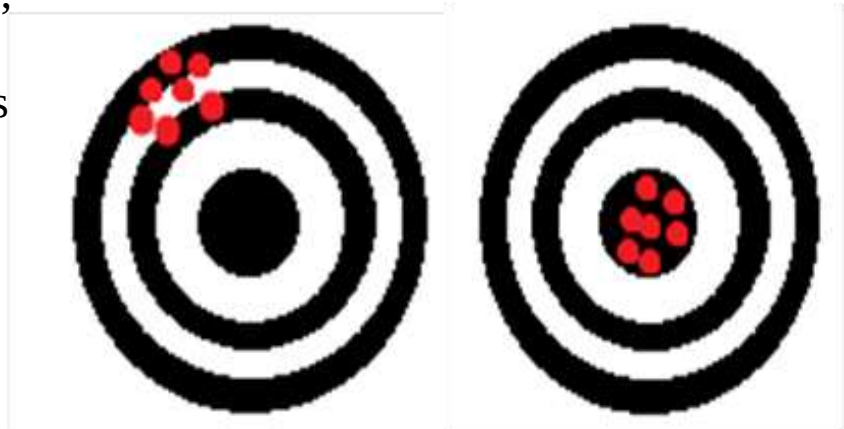
Fig 4

High *Cp* / Low *Cpk* High *Cp* / High *Cpk*

Interpreting C_{pk} :-

"C_{pk} is an **index** (a simple number) which **measures** how close a process is running to its specification limits, relative to the natural variability of the process. The **larger the index**, the less likely it is that any item will be outside the specs."

Example: "If you hunt our **shoot targets** with bow, darts, or gun try this analogy. If your shots are **falling in the same spot** forming a good group this is a **high C_p**, and when the **sighting is adjusted** so this **tight group of shots is landing on the bulls-eye**, you now have **a high C_{pk}**."

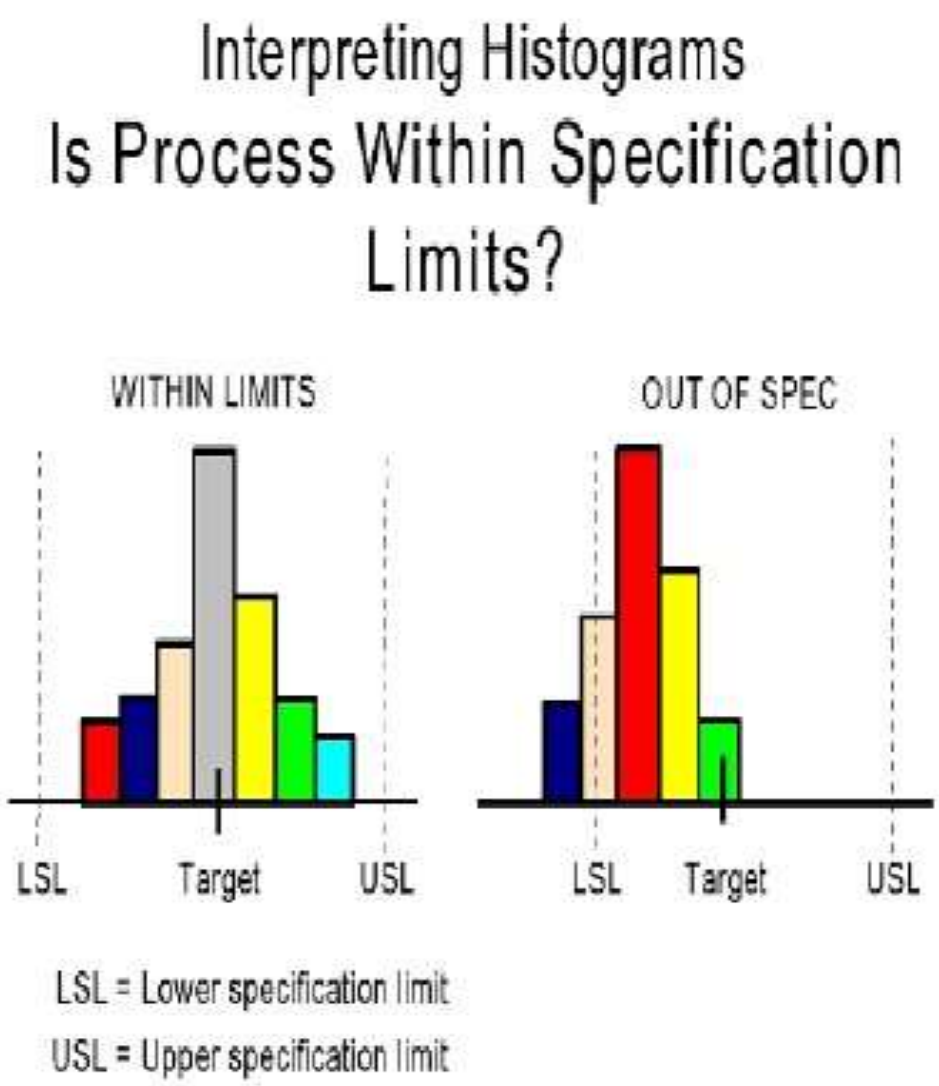


"You must have a C_{pk} of 1.33 [4 sigma] or higher to satisfy most customers."

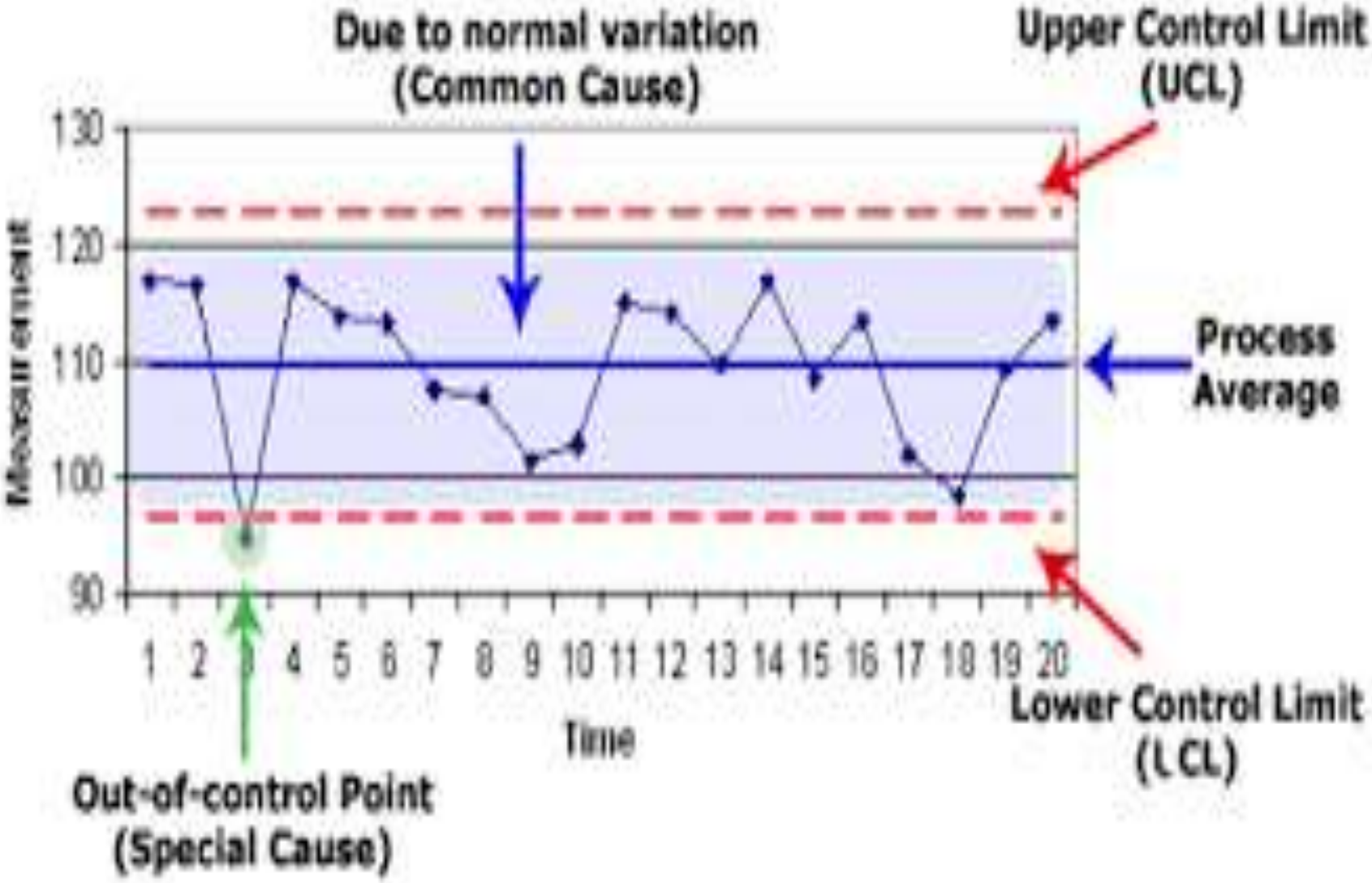
-: Histogram :-

Benefits:

- Allows you to **understand** at a glance the variation that exists in a process
- The **shape** of the histogram will show process behavior
- Often, it will tell you to **dig deeper** for otherwise unseen causes of variation.
- The shape and size of the dispersion will help **identify** otherwise hidden sources of variation
- Used to **determine** the capability of a process
- **Starting point** for the improvement process



-: 6 Quality tool: Control Chart :-



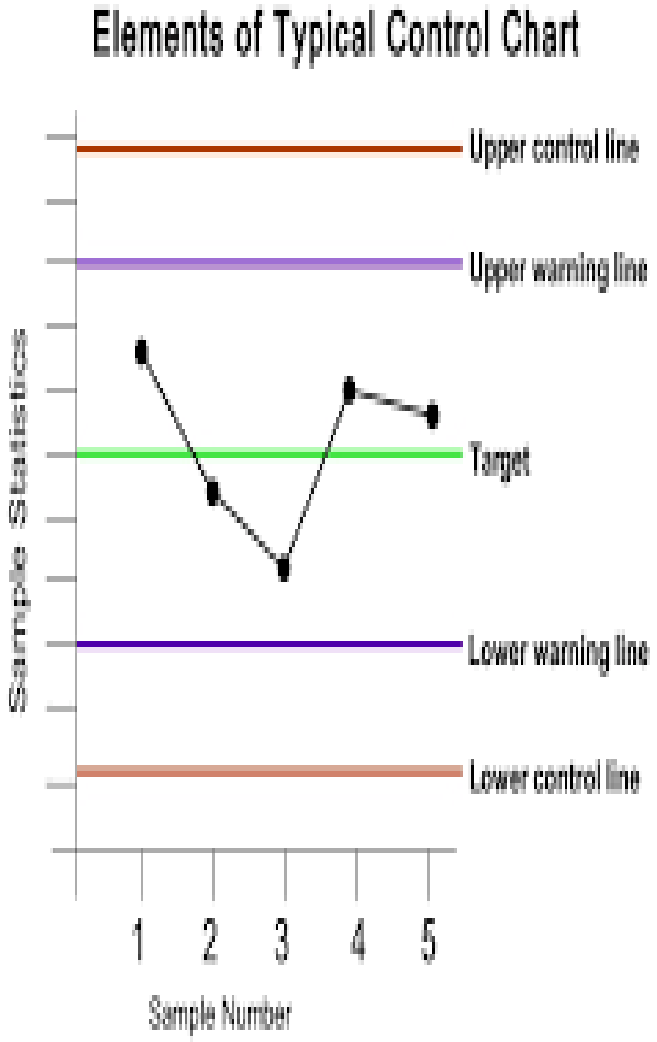
-: Control Chart :-

- **Purpose:-**

  The control chart is a graph used to study how a **process changes over time**.

- **Guidelines:-**

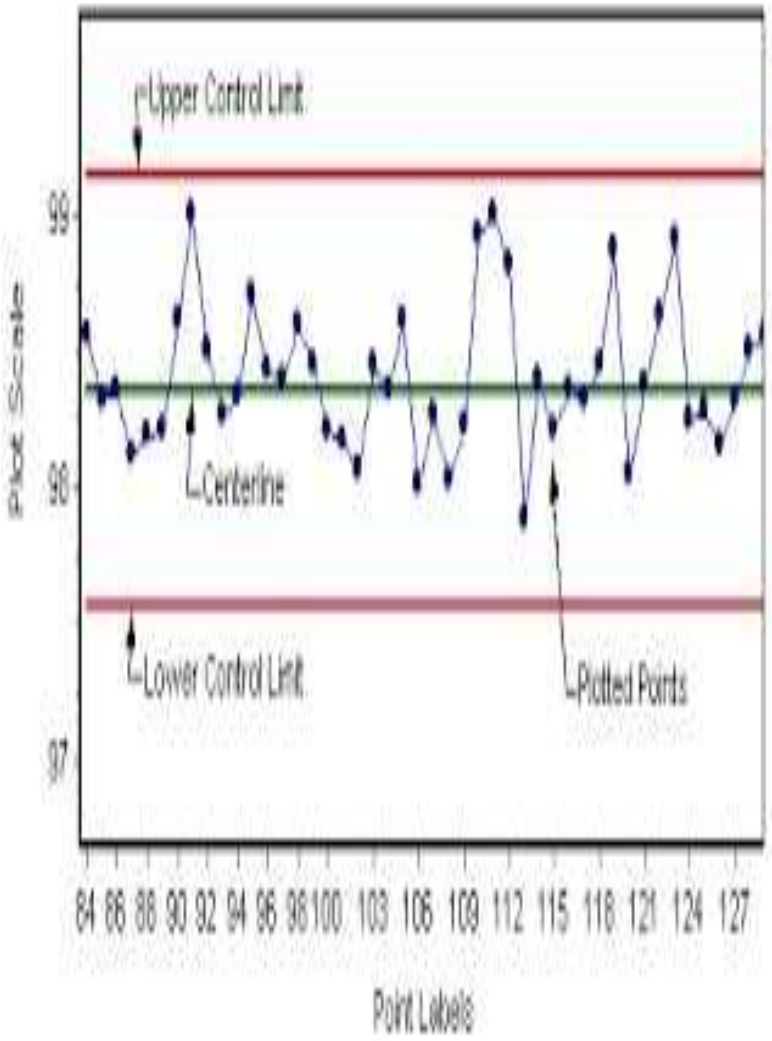
  A control chart always has a **central line** for the average, an **upper line** for the upper control limit and a **lower line** for the lower control limit. These lines are determined from historical data. By comparing current data to these lines, you can **draw conclusions** about whether the process variation is consistent (in control) or is unpredictable (out of control, affected by special causes of variation).



-: Control Chart :-

➢ **When to Use a Control Chart :-**

-When **controlling ongoing processes** by finding and correcting problems as they occur.
-When **predicting** the expected range of outcomes from a process.
- When **determining** whether a process is stable (in statistical control).
-When **analyzing** patterns of process variation from **special causes** (non-routine events) or **common causes** (built into the process).
-When **determining** whether your quality improvement project should aim to prevent specific problems or to make fundamental changes to the process.



-: Control Chart :-

➢ **Control Chart Basic Procedure :-**

1. Choose the appropriate control chart for your data.

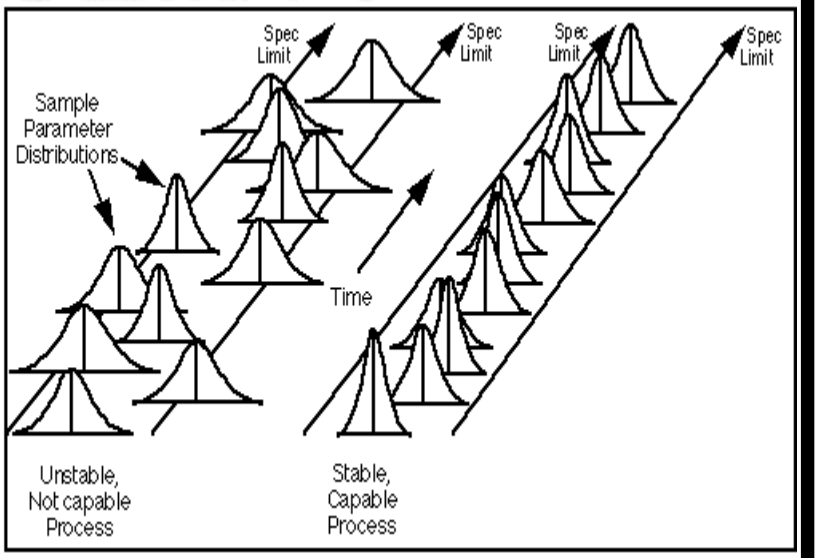
2.Determine the appropriate time period for collecting and plotting data.

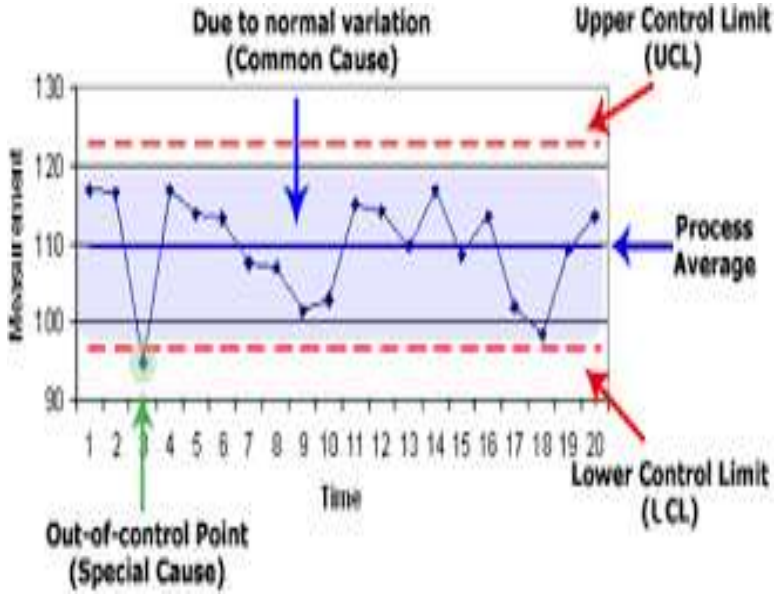
3.Collect data, construct your chart and analyze the data.

4.Look for "**out-of-control signals**" on the control chart. When one is identified, **mark it** on the chart and **investigate** the cause. ***Document how you investigated, what you learned, the cause and how it was corrected.***

5.Continue to **plot data** as they are generated. As each new data point is plotted, check for new out-of-control signals.

6 Control chart.xlsx





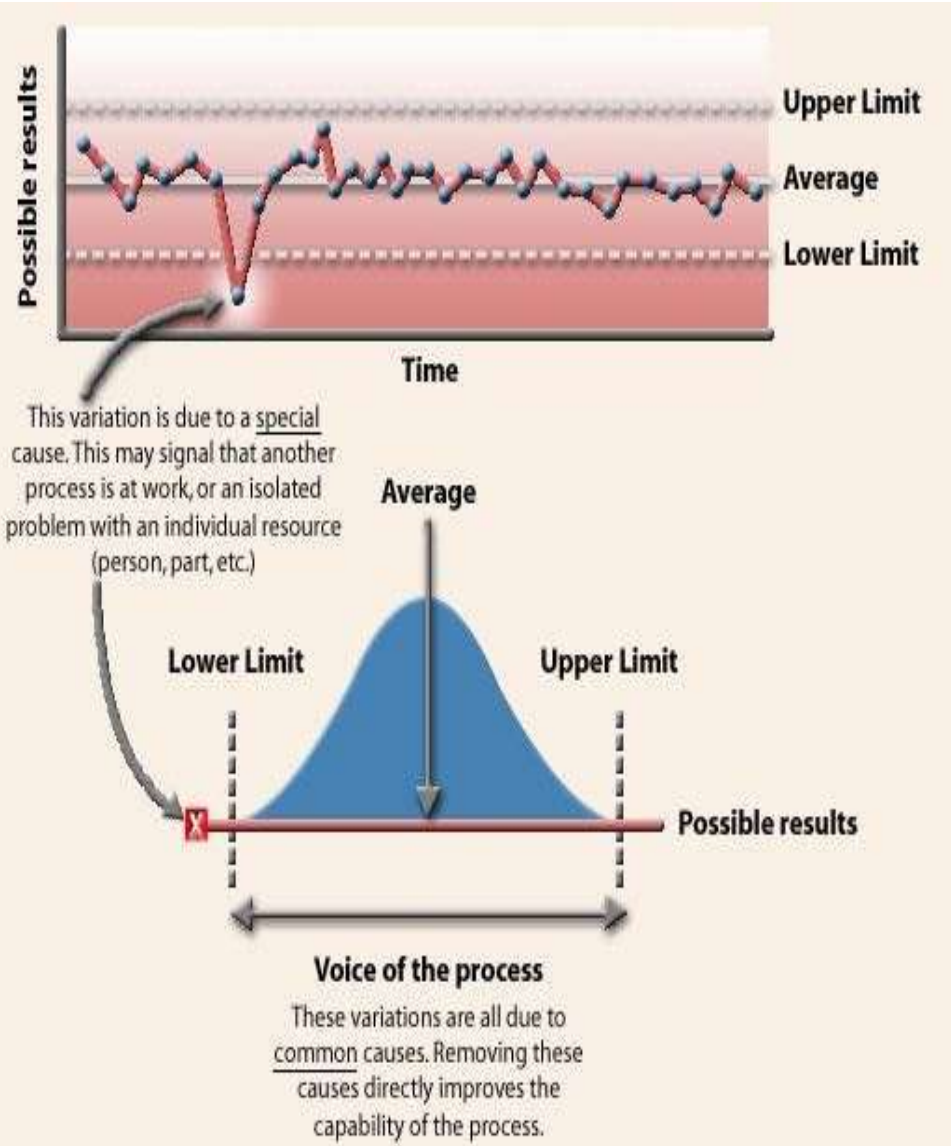
-: Control Chart :-

- **Strategy for eliminating assignable/Special cause (i.e. unpredictable errors) variation:**

- **Get timely data** so that you see the effect of the **assignable cause** soon after it occurs.
- As soon as you see something indicates that an **assignable cause of variation** has happened, search for the cause.
- **Change tools** to compensate for the assignable cause.

- **Strategy for reducing common-cause (i.e. Predictable errors) variation:**

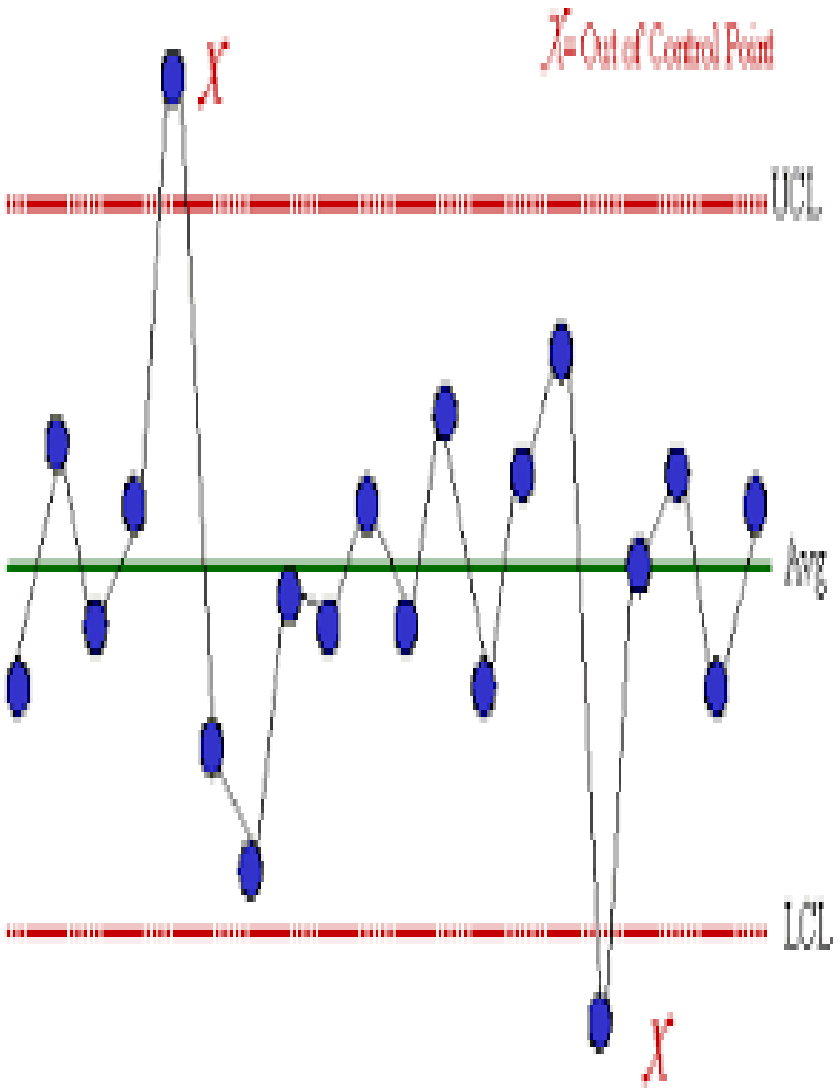
- **Reducing common-cause variation** usually requires making fundamental changes in your process
- **Addressing** the common cause variation will improve the process performance.



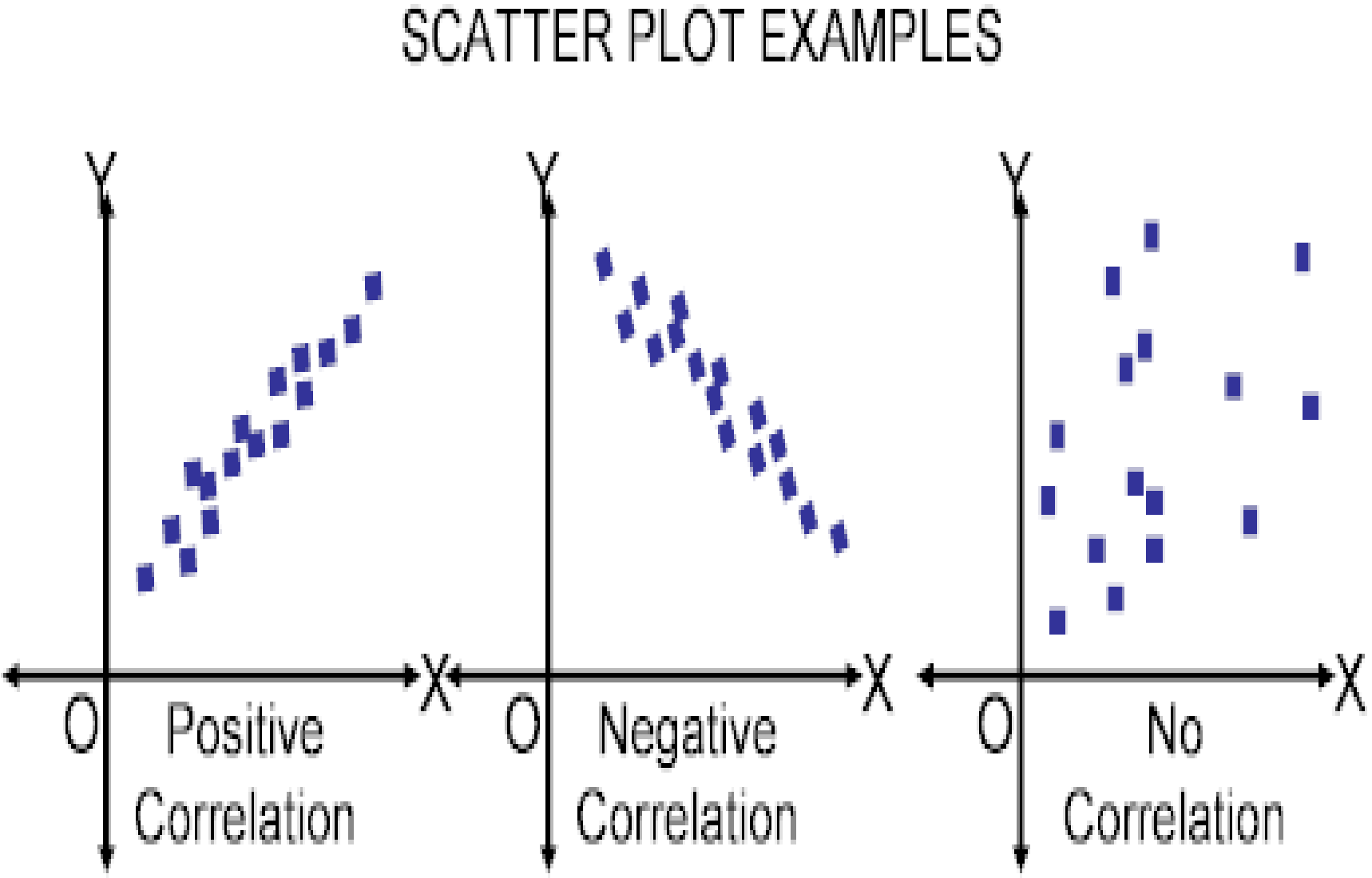
-: Control Chart :-

Benefits:

- **Predict process** out of control and out of specification limits
- **Distinguish** between specific, identifiable causes of variation
- Can be **used** for statistical process control
- Control charts allow operators to detect manufacturing problems before they occur, this **greatly reduces** the need for **product rework** or **additional product expenditures**.
- Control charts serve as the **early warning detection system**, telling you that now is the time to go in and make a change.
- After analyzing a control chart, operators need to **determine** whether to "**do something**" (i.e. adjust a behavior in the process) or "**do nothing**," (i.e. let the process run as is).



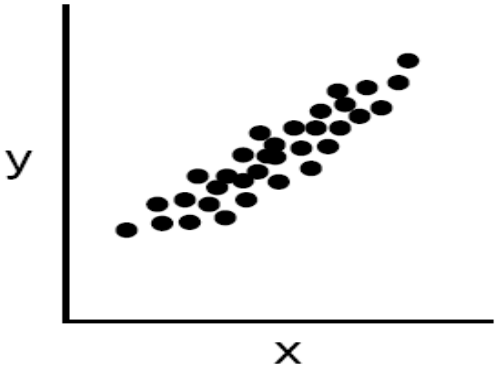
-: 7th Quality tool: Scatter Diagram :-



-: Scatter Diagram :-

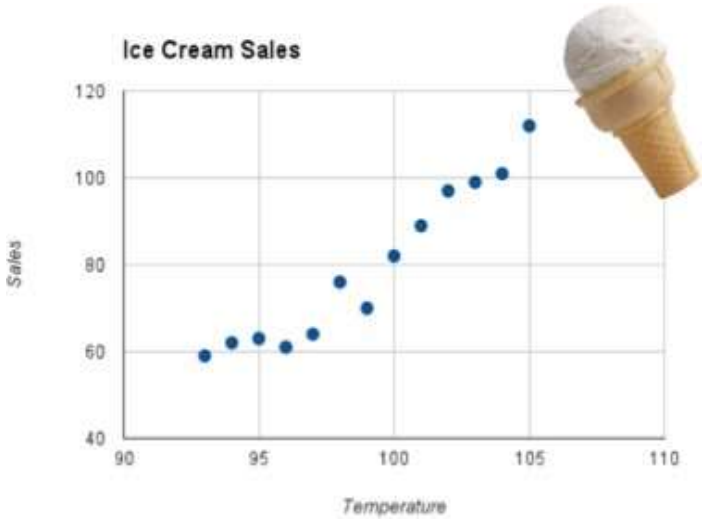
Purpose:

To **identify** the **correlations** that might exist between a quality characteristic and a factor that might be driving it



A scatter diagram **shows the correlation** between two variables in a process. These variables could be a Critical To Quality (CTQ) characteristic.

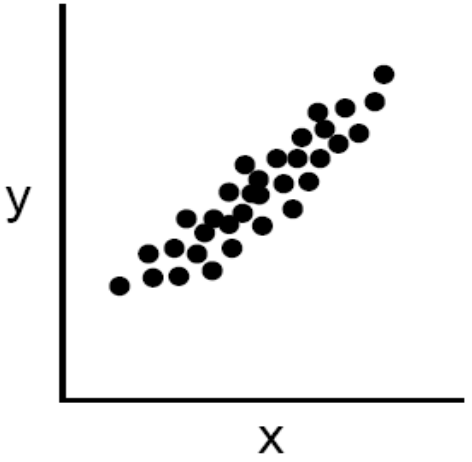
Dots representing data points are *scattered* on the diagram.



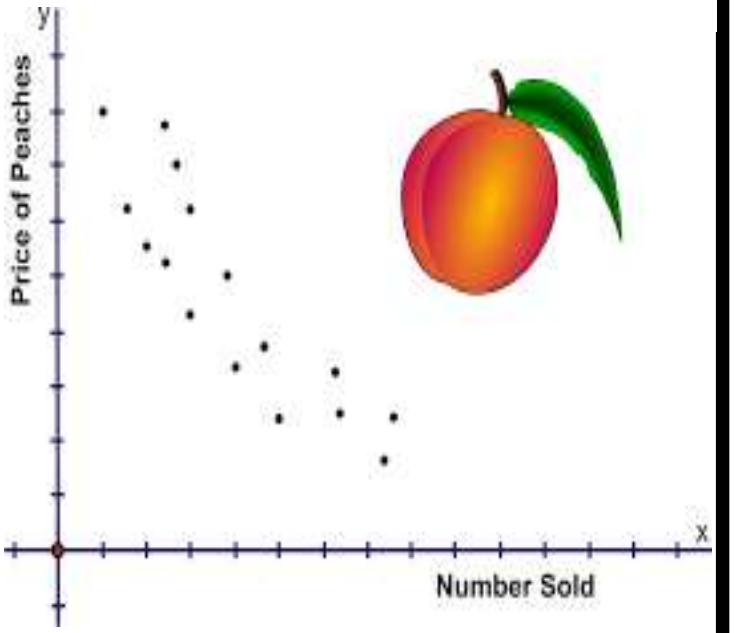
-: Scatter Diagram :-

Procedure: How is it done?

- **Decide** which paired factors you want to examine. Both factors must be measurable on some incremental linear scale.
- **Collect 30 to 100 paired data points**.
- **Find the highest and lowest value** for both variables.
- **Draw the vertical (y) and horizontal (x)** axes of a graph.
- **Plot the data**
- **Title the diagram**



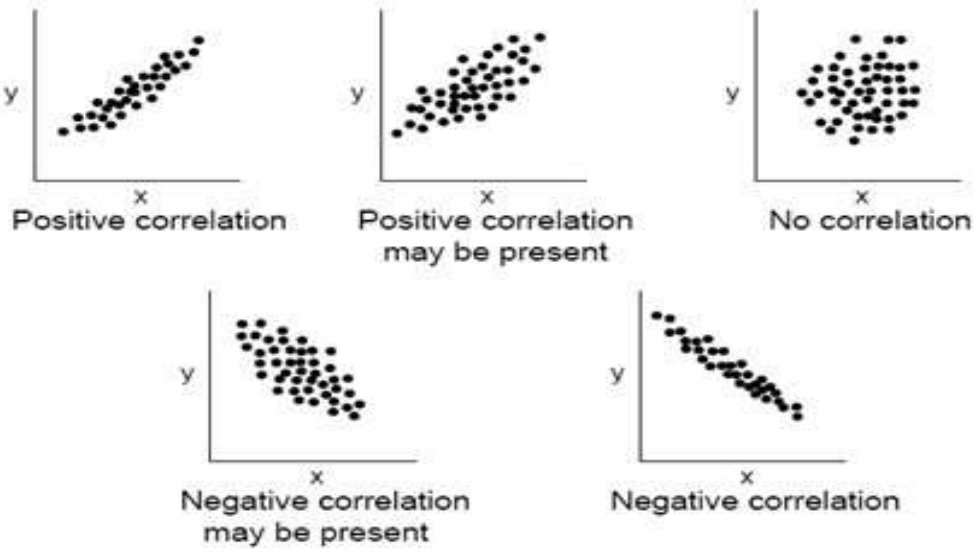
7 scatter_plots.xlsx

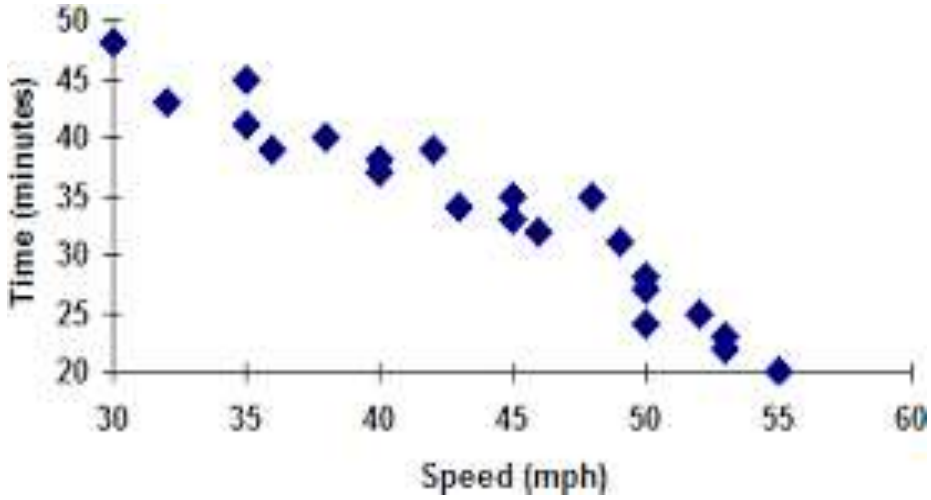


*The **shape** that the cluster of dots takes will tell you something about the **relationship** between the **two variables** that you tested.*

-: Scatter Diagram :-

- If the **variables** are **correlated**, when <u>one changes the other probably also changes</u>.
- **Dots** that look like they **are trying to form a line** are <u>strongly correlated.</u>
- Sometimes the scatter plot may show **little correlation** when all the data are considered at once.





Seven New Management and Planning Tools

A team researched and developed the seven new quality control tools, often called the seven management and planning (MP) tools, or simply the seven management tools. Not all the tools were new, but their collection and promotion were. The seven MP tools, listed in an order that moves from abstract analysis to detailed planning, are:

1. Affinity diagram
2. Interrelations diagram
3. Tree diagram
4. Matrix diagram
5. Matrix data analysis
6. Arrow diagram
7. Process decision program chart (PDPC)

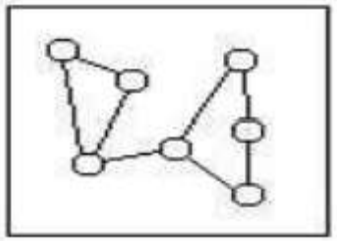
1. Affinity Diagram

The purpose of an affinity diagram is to provide a visual representation of grouping of a large number of ideas or factors or requirements into logical sets of related items to help one organize action plans in a systematic manner.



2. InterRelations diagram:

shows cause-and-effect relationships and helps you analyze the natural links between different aspects of a complex situation.



3. Tree Diagram

The purpose of the tree diagram is to explore ways and means to achieve an objective, develop a list of alternate means to reach the desired situation in a sequential order and to present them in a visual form.



4. Matrix Diagram

Purpose

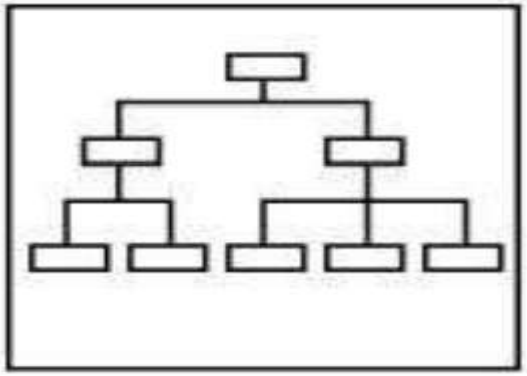
The purpose of a matrix diagram is to explore the existence and the extent of relations between individual items in two sets of factors or features or characteristics and express them in a symbolic form that is easy to understand. The purpose for which the tool is most frequently used is to understand the relation between customer expectations as expressed by the customers and product characteristics as designed, manufactured and tested by the manufacturer.

5. Matrix Data Analysis Diagram

Purpose

The purpose of matrix data analysis diagram is to present numerical data about two sets of factors in a matrix form and analyze it to get numerical output. The factors most often are products and product characteristics. The purpose then is to analyse the data on several characteristics for a number of products and use the information to arrive at optimum values for the characteristics for a new product or to decide the strong points of a product and use the information for designing a strategy for the promotion of the product

6. Arrow diagram:
shows the required order of tasks in a project or process, the best schedule for the entire project, and potential scheduling and resource problems and their solutions.



7. **Process decision program chart (PDPC):** systematically identifies what might go wrong in a plan under development.

